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College of Arts and Sciences

NORSU Bais Consultlink: Instructor – Student Consultation Scheduling and Notification System for CAS Department

A Capstone Project Presented to the Faculty of the Computer Science and Information Technology Department College of Arts and Sciences Negros Oriental State University Bais City

In Partial Fulfillment of the Requirements for the degree of Bachelor of Science and Information Technology

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CHAPTER 1

Project Context

At NORSU, consultation is an important part of the academic process, allowing students to seek guidance, clarification, and academic support from their instructors. However, the current method of consultation is highly informal and unorganized. Most students simply approach their instructors without prior notice, resulting in unexpected interruptions overlapping consultations, and difficulty accommodating all students.

On the instructor's side, many consultations are not properly recorded or tracked. Since most instructors rely on verbal arrangements, handwritten notes, or memory, consultation schedules are often forgotten, misplaced, or not updated. This leads to confusion, missed appointments, and inconsistent documentation of student concerns.

Students also lack a reliable way of knowing whether the instructor is available. They often go directly to the faculty office hoping to find their instructor, only to discover that the instructor is in class, in a meeting, or not around. This results in wasted time and delays in obtaining academic assistance.

To address these issues, the researchers propose ConsultLink, a web-based Instructor Consultation Scheduling and Notification System. This system will allow students to conveniently book consultation appointments online and receive real-time updates. Instructors will also receive email notifications whenever a student schedules a consultation, ensuring they are aware of upcoming appointments.

With automated scheduling, digital records, and instant notifications, ConsultLink aims to create a more organized, efficient, and transparent consultation process. The proposed system will help reduce interruptions improve communication, support academic productivity, and ensure better time management for both instructors and students.

Purpose and Description

The primary purpose of this project is to develop ConsultLink: Instructor Student Consultation Scheduling and Notification System. This web-based platform is meticulously designed to revolutionize the consultation process at College of Arts and Sciences in Negros Oriental State University (NORSU) Bais City. By transitioning from the current unstructured,

walk-in consultation model to a streamlined and automated system, ConsultLink aims to provide tangible benefits for instructors, students, and the institution as a whole.

Online Appointment Booking: Students gain the ability to book consultation appointments online through an intuitive and user-friendly interface. This feature eliminates the need for impromptu visits to instructors' offices, reducing interruptions and increasing convenience. Students can view available time slots, select their preferred time, and provide a brief description of the consultation topic.

Instructor Schedule Management: Instructors are empowered with a comprehensive schedule management system. They can digitally view incoming consultation requests, approve or decline appointments based on their availability, and block out specific times for other commitments. This functionality ensures that instructors maintain control over their schedules and can effectively manage their consultation workload.

The system incorporates an automated email notification system. When a student schedules a consultation, the instructor receives an immediate email notification with details of the appointment. Reminders are also sent closer to the consultation time, minimizing the risk of missed appointments.

Students can log in to the system and view the status of their consultation requests. They can see whether their request is pending, approved, or declined, ensuring transparency and reducing uncertainty.

Admin Dashboard (Future Implementation): While not explicitly mentioned in the original document, a potential future enhancement would be the inclusion of an administrative dashboard. This would allow administrative staff to oversee the entire consultation process, generate reports on consultation activity, manage user accounts, and configure system settings.

The purpose of this project is to develop ConsultLink: Instructor-Student Consultation Scheduling and Notification System, a web-based platform designed to streamline the consultation process at NORSU. The system aims to replace the unstructured walk-in approach with a more organized and automated scheduling system. The instructors will be providing a reliable schedule, automated notifications, and digital records of consultations.

The students offering a convenient way to book appointments anytime and ensuring the instructor is available. The Institution promoting a more organized and efficient academic

support system. ConsultLink, consultation sessions will become more systematic, time-efficient, and well-documented, improving communication and academic productivity.

Objectives of the Study

To develop a web-based consultation scheduling and notification system that enhances the communication and appointment process between instructors and students at NORSU. The main objective of this study is to develop NORSU Bais Consultlink an efficient and user-friendly Instructor–Student Consultation Scheduling and Notification System designed to streamline and enhance the consultation process within the CAS Department at NORSU Bais Campus. The system aims to improve communication, reduce scheduling conflicts, and provide a more organized and accessible method for managing academic consultations.

Specific Objectives

1. To create a centralized online scheduling platform where instructors can set consultation hours and students can conveniently book appointments.
2. To implement an automated notification feature that alerts users about approved, pending, cancelled, or upcoming consultations.
3. To provide real-time updates and status tracking to help both instructors and students stay informed about consultation requests.
4. To build a secure database that stores consultation history, user information, and scheduling logs for academic reference.
5. To design a user-friendly and mobile-responsive interface that ensures accessibility across different devices.
6. To allow instructors to manage and modify their schedules easily, including adding, updating, or disabling consultation slots.
7. To enable students to view instructor availability and choose consultation times that best fit their schedules.
8. To include a notification system for reminders to reduce missed appointments and improve punctuality.
9. To generate reports and summaries regarding consultation frequency, attendance, and scheduling trends for administrative evaluation.
10. To minimize manual processes, such as walk-ins and paper-based scheduling, by transitioning to an automated digital system.

11. To improve communication flow between instructors and students by providing a reliable and structured consultation channel.
12. To support academic performance and student success by making consultations more organized, accessible, and consistent.

Scope and Limitations

Scope

The system will include features such as online consultation booking, instructor schedule management, email notifications, and digital record-keeping. It will support two types of users the student who can request, view, and track consultation schedules and also the instructors can view, manage, and receive notifications of consultation requests. The development of NORSU Bais ConsultLink focuses on creating a digital platform that streamlines and enhances the consultation scheduling process between instructors and students within the College of Arts and Sciences (CAS) Department of Negros Oriental State University–Bais Campus.

Limitations

The development of the NORSU Bais ConsultLink system is limited only to the College of Arts and Sciences (CAS) of NORSU Bais Campus and does not extend to other departments or campuses. The system also relies heavily on stable internet connectivity. The students and instructors with weak or unstable connections may experience delays in accessing features or receiving updates. The accuracy of consultation schedules depends on instructors manually updating their availability, and the system cannot automatically detect changes unless entered by the user. In the notification feature may be affected by third party email or SMS services, which can result in delayed or failed message delivery. The system does not provide automated academic advising or generate consultation recommendations, as it only facilitates scheduling and reminders. It also offers limited conflict detection, preventing only double-booking within the platform but not conflicts with external schedules such as classes or meetings. Lastly, the system focuses solely on scheduling and notifications and does not include built-in video conferencing or real-time chat features, requiring users to use external platforms for online consultations.

CHAPTER 2

Related Literature

Based on the study of Harper and Mayorga (2020), the project entitled "Automated Academic Scheduling Platform" discusses the growing need for efficient digital scheduling systems in educational institutions. The authors emphasize that traditional manual scheduling methods often lead to double bookings, miscommunication, and delays in academic consultations. Harper and Mayorga argue that implementing an automated consultation scheduling system can significantly improve coordination between instructors and students by offering real-time availability, reducing waiting times, and minimizing scheduling conflicts. This is particularly relevant to the development of ConsultLink, as many schools still rely on outdated methods such as walk-ins or manual logbooks that hinder productivity and cause unnecessary frustration among users. Incorporating a modern digital platform aligns with the findings of Harper and Mayorga, presenting an opportunity to enhance both academic workflow and user convenience.

The study of Lopez and Hernandez (2021) titled "Digital Academic Advising in Contemporary Higher Education" highlights the importance of accessible and well-structured instructor–student consultation processes. Their research shows that improved advising systems lead to better academic performance, student engagement, and retention. Lopez and Hernandez explain that students benefit from platforms that allow easy scheduling, timely responses, and structured academic support. This supports the purpose of ConsultLink, which aims to replace outdated consultation practices with a digital scheduling and notification system that promotes organized, responsive, and efficient communication.

The study of Ramos and Choi (2022), entitled "Technology-Mediated Student–Faculty Interaction in Universities," examines the role of consistent communication in fostering student academic success. Their findings reveal that regular consultations improve motivation, enhance understanding of course content, and strengthen a student's sense of academic belonging. This literature directly relates to ConsultLink, which provides a structured platform for booking consultations, tracking meeting histories, and facilitating communication. By addressing the challenges identified by Ramos and Choi, the system enhances the consultation experience and supports academic performance.

The study by Nguyen, Patel, and Brooks (2021) titled "Effectiveness of Automated Notification Systems in Reducing Appointment No-Shows" examines how digital reminders improve attendance rates in academic settings. Their analysis found that automated SMS and digital notifications significantly reduce missed appointments and increase user engagement. These findings highlight the importance of integrating timely alerts in a scheduling system. ConsultLink benefits from this approach, as many students overlook scheduled consultations due to workload or lack of reminders. Automated notifications through SMS, email, or in-app alerts ensure better reliability in consultation attendance.

The study of Norman and Nielsen (2021) titled "User-Centered Design Principles for Modern Digital Platforms" emphasizes the importance of intuitive, accessible, and user-friendly system design. Their research shows that platforms with clear navigation and simple interfaces are more likely to be adopted by users with varying technology skills. Poorly designed systems result in decreased engagement and user frustration. This has direct implications for ConsultLink, which must integrate user-centered design features such as easy scheduling options, accessible notification modules, and a simple interface to ensure a positive user experience.

The study by Leong and Lovatt (2021) entitled "Enhancing Academic Advising Services Through Automated Scheduling Systems" investigates how appointment platforms improve academic support services. Their research indicates that automated scheduling reduces instructor workload, enhances transparency, and increases flexibility in appointment selection. These findings support the development of ConsultLink, especially in institutions where faculty balance teaching, administrative duties, and student consultations. Digital scheduling streamlines their workflow and improves advising quality.

The study by Selwyn and Gouseti (2022) titled "Education and Technology in the Post-Digital Academic Environment" provides a broader perspective on incorporating digital systems in educational institutions. They argue that technology adoption must address institutional needs, user expectations, and practical workflow problems. Institutions that fail to provide accessible and well-designed systems often experience resistance from users. This literature aligns with ConsultLink, which aims to modernize the consultation workflow and provide an accessible and relevant digital tool that improves academic communication and coordination.

Related Studies

According to Santos (2020) in the study "Faculty Consultation Management System for Academic Institutions," the lack of automated scheduling systems causes overlapping schedules, missed appointments, and inefficient consultation tracking. Santos emphasizes that an online scheduling platform allows students to check instructor availability, request appointments, and receive timely updates. These findings align with ConsultLink's goal of centralizing and modernizing consultation processes.

Dela Cruz and Ramirez (2021) developed an "Automated Student–Faculty Consultation Appointment System" to solve communication delays between students and instructors. Their study found that automated notifications—such as SMS and email alerts—significantly reduced the number of missed consultations. This supports ConsultLink's notification module, which is designed to improve accountability and time management.

In the research of Wong et al. (2020) titled "University Student Support Systems: Enhancing Academic Communication Through Technology," digital platforms were shown to increase student engagement in academic support services. The authors found that technology-enabled consultation systems help students who feel hesitant to consult face-to-face. These insights reinforce the importance of designing ConsultLink with an intuitive interface to encourage consistent usage.

The study by Villanueva and Lee (2022) entitled "Mobile-Based Appointment Systems for Academic Consultation" explored the impact of mobile accessibility in university consultation processes. Their results show that mobile-compatible systems improve scheduling flexibility, reduce waiting times, and enhance consultation records. These findings support ConsultLink's accessibility across different devices.

Gomez (2020) in the study "Integrating Digital Tools for Faculty Workload Management," highlights the need for systematic consultation logs that help instructors monitor student concerns, academic progress, and recurring issues. Gomez's findings suggest that digital records enhance the quality of academic advising by providing historical data that instructors can reference during sessions. This aligns with ConsultLink's record-tracking component, which aims to organize consultation histories for future academic evaluation.

From the study by Navarro et al. (2021) titled "Academic Advising Systems and Their Role in Student Success," it is emphasized that efficient consultation processes are directly linked to higher student retention and academic performance. They recommend universities

adopt digital systems that streamline advising workflows and reduce response times. Their findings support ConsultLink as a structured academic consultation platform.

THEORITICAL FRAMEWORK

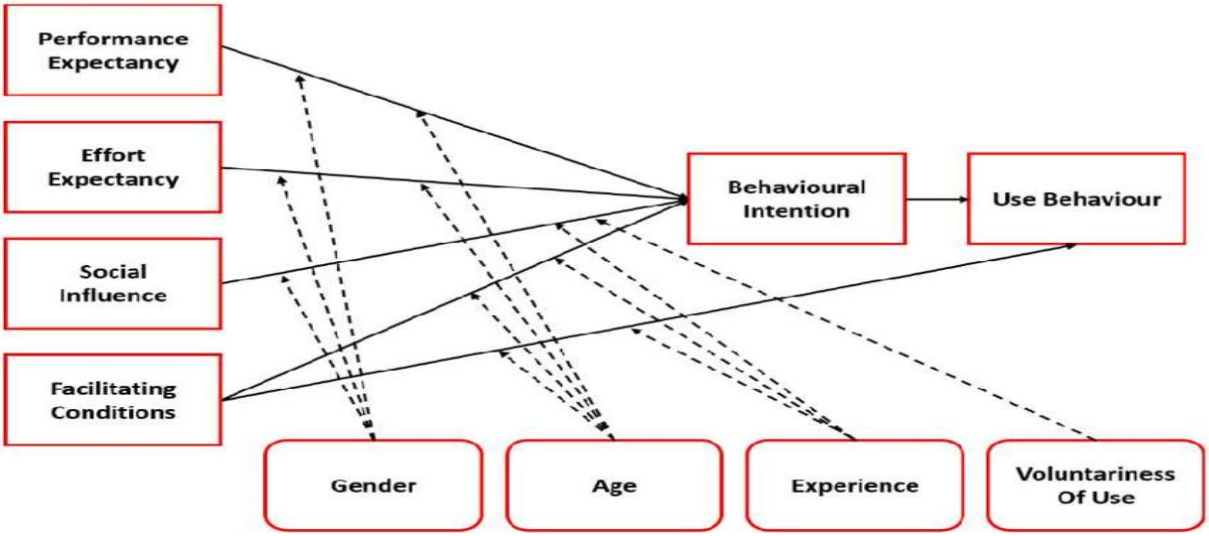

Figure 1: UTAUT MODEL

The development of ConsultLink: Instructor–Student Consultation Scheduling and Notification System is anchored in the Unified Theory of Acceptance and Use of Technology (UTAUT). This framework is appropriate for the study because it provides a comprehensive explanation of how users, both instructors and students decide to adopt and utilize new digital systems. The success of ConsultLink depends largely on whether its target users perceive it as beneficial, easy to use, and supported within the academic environment.

UTAUT identifies four key determinants that directly relate to the system. Performance Expectancy is addressed through the system's ability to streamline scheduling and improve communication efficiency. Effort Expectancy is reflected in the user-friendly design of ConsultLink, which ensures that even users with limited technical skills can easily navigate the platform. Social Influence plays a role as instructors, peers, and institutionalpolicies encourage the adoption of the system. Lastly, Facilitating Conditions are met through the availability of technical support and infrastructure necessary for system operation.

By aligning the study with the UTAUT model, it establishes a strong theoretical basis for analyzing user acceptance, adoption behavior, and overall system effectiveness. Thus, UTAUT does not merely provide a definition but serves as the guiding theory that supports the study's objectives of measuring usability, acceptance, and the potential adoption of ConsultLink within the academic setting.

Conceptual Framework

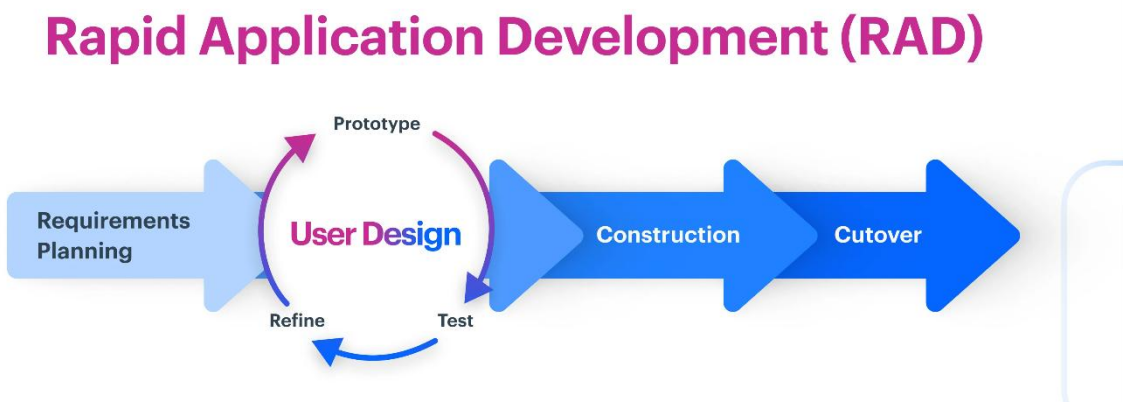

Figure 2: Rapid Application Development

The development of ConsultLink follows the Rapid Application Development (RAD) model, which is selected for its emphasis on iterative development, fast delivery, and intensive user involvement. RAD is particularly suitable for this project because the system requires continuous refinement based on real user feedback to ensure that its features, including scheduling, notifications, and consultation tracking, meet the practical needs of instructors and students.

The RAD model consists of four phases: Requirements Planning, User Design, Construction, and Cutover. In this study, the Requirements Planning phase allows the researchers to identify the essential features of the system. The User Design phase is critical, as both students and instructors collaborate with the developers to shape and refine the design through multiple prototypes. This ensures that the interface remains intuitive and aligned with user expectations. The Construction phase focuses on rapid development of system modules, while the Cutover phase prepares the system for deployment and user training.

RAD is chosen not only for its speed but also for its structured approach that minimizes project delays and prevents scope creep—an important consideration for Capstone development. By using RAD, the system can be tested, evaluated, and improved in shorter cycles, ensuring a more efficient development process and a final product that is highly responsive to user needs. Therefore, RAD supports the study by providing a practical and user-centered framework for building a functional and effective consultation scheduling system.